

Project Title

ParkEase: Integrated Parking & Vehicle Services Management System

1. Project Overview

ParkEase is a web-based software solution designed to digitize the day-to-day operations of a public parking facility that also offers tyre clinic and battery hire services. The system will support accurate record keeping, automated fee calculation, receipt generation, and daily revenue reporting.

2. Problem Statement

Parkville currently manages parking, tyre services, and battery hire manually, which leads to poor record keeping, revenue leakages, and inefficiencies. There is a need for a centralized digital system to manage vehicle registration, service transactions, and daily reporting.

3. Project Scope

Core Module (Mandatory)

- Parking Management System

Other Services (Extra Credit)

- Tyre Clinic Management
- Battery Hire & Sales Management

Note: Students are required to fully implement the Parking Management module. Optional modules earn additional marks if implemented correctly.

4. User Roles

Parking Attendant

- Register vehicles on arrival
- Issue parking receipts
- Sign out vehicles

- Record service transactions
- Set service or hire prices

System Admin

- View daily reports
- Manage users

5. Functional Requirements

5.1 Vehicle Registration

The system shall allow a user to register a vehicle by capturing: - Driver name - Vehicle type (Truck, Personal Car, Taxi, Coaster, Boda-boda) - Number plate - Vehicle model and color - Arrival time and date - Phone number - NIN number (for boda-bodas)

5.2 Parking Charges

The system shall automatically calculate parking fees based on vehicle type and duration:

- Trucks: UGX 5,000 (day), UGX 10,000 (night), UGX 2,000 (<3 hours)
- Personal Cars: UGX 3,000 (day), UGX 2,000 (night), UGX 2,000 (<3 hours)
- Taxis: UGX 3,000 (day), UGX 2,000 (night), UGX 2,000 (<3 hours)
- Coasters: UGX 4,000 (day), UGX 2,000 (night), UGX 3,000 (<3 hours)
- Boda-bodas: UGX 2,000 (day/night), UGX 1,000 (<3 hours)

Day time: 6:00am – 6:59pm

Night time: 7:00pm – 5:59am

5.3 Receipt Management

- The system shall generate a unique receipt number for every transaction.
- Receipts shall be used during vehicle sign-out.

5.4 Sign-Out Process

On sign-out, the system shall record: - Receiver name - Receipt number - Date and time - Phone number - Gender - NIN number

5.5 Reporting

The system shall: - Display all signed-out vehicles for a selected day - Display total daily revenue per section (Parking, Tyre, Battery)

6. Other Services

6.1 Tyre Clinic

- Predefined tyre prices by size and model
- Services: pressure (UGX 500), puncture fixing (UGX 5,000), valves (UGX 5,000)

6.2 Battery Section

- Battery hire and sales
- Prices set by section manager

7. Validation Rules

- Names must start with a capital letter and contain no numbers
- Number plates must be alphanumeric, start with 'U', and be less than 6 characters •
- Phone numbers must be valid Ugandan numbers
- NIN numbers must follow a valid format
- Required fields must not be empty
- Validation must be implemented on both frontend and backend

8. Technology Requirements

This project is applicable to both JavaScript and Python CSE tracks.

Frontend (Both Tracks)

- HTML & CSS (User Interface)
- JavaScript (Client-side logic and validation)

Backend (Choose Based on Track)

JavaScript Track - Node.js & Express

Python Track - Python and Django

Database (Both Tracks)

- SQLite

Other Requirements (Both Tracks)

- Git & GitHub (Version control)

9. Deliverables

- Fully functional web application
- GitHub repository with meaningful commits
- Link to completed project
- README documentation (setup, features, assumptions)
- 5- m i n u t e project demo

10. Assessment Criteria

- Functionality & correctness – 40%
- Code quality & structure – 20%
- Validation & error handling – 15%
- UI/UX & usability – 10%
- Presentation – 10%
- Sign in/Log in Authentication – 5%

11. Project Duration

- Total duration: 6 weeks
- Weekly progress check-ins recommended

Good luck, and build with purpose!

